Solution 2 and 3

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Remark:If you find any mistakes, typos, or have any other feedback, please let me know.

1 Continuous-time stochastic processes

Problem 1

On $(\Omega, \mathcal{F}, \mathbb{P})$, let $X = \{X_t : t \geq 0\}$ be a stochastic process that is measurable and T is a random time. Prove that

(a) X_T is a random variable if T is finite;

(b) All sets of the form $\{X_T \in A\}$ and $\{X_T \in A\} \cup \{T = \infty\}$ with $A \in \mathscr{B}(\mathbb{R}^d)$ forms a σ-algebra.

Proof. (a) Since X is measurable, the map $X : \Omega \times [0, \infty) \to \mathbb{R}^d$ is $(\mathcal{F} \otimes \mathscr{B}([0, \infty)), \mathscr{B}(\mathbb{R}^d))$-measurable. For any $A \in \mathscr{B}(\mathbb{R}^d)$, the set $S_A = \{(\omega, t) \in \Omega \times [0, \infty) : X_t(\omega) \in A\}$ belongs to the product σ-algebra $\mathcal{F} \otimes \mathscr{B}([0, \infty))$. Since T is a random time and $T < \infty$, $T : \Omega \to [0, \infty)$ is $(\mathcal{F}, \mathscr{B}([0, \infty)))$-measurable. Consider $Z : \Omega \to \Omega \times [0, \infty)$ defined by $Z(\omega) = (\omega, T(\omega))$. The components of Z are:

- $\pi_1 \circ Z(\omega) = \omega$, which is $(\mathcal{F}, \mathcal{F})$-measurable.
- $\pi_2 \circ Z(\omega) = T(\omega)$, which is $(\mathcal{F}, \mathscr{B}([0, \infty)))$-measurable.

Therefore, Z is $(\mathcal{F}, \mathcal{F} \otimes \mathscr{B}([0, \infty)))$-measurable.

$$\{\omega : X_T(\omega) \in A\} = \{\omega : (\omega, T(\omega)) \in S_A\} = Z^{-1}(S_A)$$

Since Z is measurable and $S_A \in \mathcal{F} \otimes \mathscr{B}([0, \infty))$, the preimage $Z^{-1}(S_A)$ must belong to $\mathcal{F}$.

$$\{X_T \in A\} \in \mathcal{F} \quad \text{for all } A \in \mathscr{B}(\mathbb{R}^d)$$

Thus, X_T is an $\mathcal{F}$-measurable random variable.

(b) Let $\mathcal{G}$ be the collection of sets of the form $\{X_T \in A\}$ and $\{X_T \in A\} \cup \{T = \infty\}$ for $A \in \mathcal{B}(\mathbb{R}^d)$. We prove $\mathcal{G}$ is a σ-algebra on Ω.

- $\Omega \in \mathcal{G}, \text{since } \Omega = \{T < \infty\} \cup \{T = \infty\} = \{X_T \in \mathbb{R}^d\} \cup \{T = \infty\} \in \mathcal{G}$.
- $\{X_T \in A\}^c = \{w : T = \infty \text{ or } T(w) < \infty, X_{T(w)}(w) \in A^c\} = \{X_T \in A^c\} \cup \{T = \infty\} \in \mathcal{G}$.
  $(\{X_T \in A\} \cup \{T = \infty\})^c = \{X_T \in A\}^c \cap \{T < \infty\} = \{X_T \in A\}^c \in \mathcal{G}$.

- For $C_i \in \mathcal{G}, i = 1, 2, \cdots$. If $\exists j$ s.t. $C_j = \{X_T \in A_j\} \cup \{T = \infty\}$, then $\cup C_i = (\cup\{X_T \in A_i\}) \cup \{T = \infty\} \in \mathcal{G}$.

□

Problem 2

Let $\{\mathcal{F}_t : t \geq 0\}$ be a filtration and X be an adapted process relative to $\{\mathcal{F}_t : t \geq 0\}$. Set $T = \inf\{t \geq 0 : X_t \in A\}$. Prove that

(a) if A is open and the sample paths of X are right-continuous, then T is an optional time of $\{\mathcal{F}_t : t \geq 0\}$.

(b) if A is closed and the sample paths of X are continuous, then T is a stopping time of $\{\mathcal{F}_t : t \geq 0\}$.

Proof. (a) For $t \geq 0$, we have

$$\begin{aligned} \{T < t\} &= \{\exists s < t : X_s \in A\} \\ &= \{\exists q < t, q \in \mathbb{Q} : X_q \in A\} \\ &= \bigcup_{q \in \mathbb{Q}, q < t} \{X_q \in A\} \in \mathcal{F}_t, \end{aligned} \tag{1}$$

where the first equality is due to the definition of infimum, and the second equality due to right continuity of paths and openness of A.

(b) For $t \geq 0$, we have

$$\begin{aligned} \{T > t\} &= \{\{X_s\}_{s \in [0,t]} \cap A = \emptyset\} \\ &= \bigcup_{n=1}^{\infty} \{\mathrm{dist}(\{X_s\}_{s \in [0,t]}, A) \geq \frac{1}{n}\} \\ &= \bigcup_{n=1}^{\infty} \bigcap_{q \in [0,t] \cap \mathbb{Q}} \{\mathrm{dist}(X_q, A) \geq \frac{1}{n}\} \in \mathcal{F}_t. \end{aligned} \tag{2}$$

The continuity of X implies that $\{X_s\}_{s \in [0,t]}$ is a compact set, and hence if it does not intersect a closed set A, it must have positive distance to A; this gives the second equality.

□

Problem 3

Let $\{X_t : t \geq 0\}$ be a progressively measurable process relative to $\{\mathcal{F}_t : t \geq 0\}$ and let T be a finite stopping time of $\{\mathcal{F}_t : t \geq 0\}$. Prove that

(a) X_T is $\mathcal{F}_T$-measurable.;

(b) the process $\{X_{T \wedge t} : t \geq 0\}$ is progressively measurable relative to $\{\mathcal{F}_t : t \geq 0\}$.

Proof. (a) Let $A \in \mathscr{B}(\mathbb{R})$. We have

$$\{X_T \in A\} \cap \{T \leq t\} = \{X_{T \wedge t} \in A\} \cap \{T \leq t\}.$$

It suffices to check that $\{X_{T\wedge t} \in A\} \in \mathcal{F}_t$.

In fact, the map $\omega \mapsto X_{T(\omega)\wedge t}(\omega)$ can be written as the composition of the two maps:

$$\omega \mapsto (t', \omega') = (T(\omega) \wedge t, \omega), \quad (t', \omega') \mapsto X_{t'}(\omega').$$

The first map is measurable from $(\Omega, \mathcal{F}_t)$ to $([0,t] \times \Omega, \mathscr{B}([0,t]) \times \mathcal{F}_t)$ by the definition of stopping times, while the second is measurable since X is progressively measurable.

(b)As above, we also use the idea that the composition of mappings preserves measurability. In this part, we omit further details here. And by the definition of progressively measurable, just need to show that

$$H : [0,S] \times \Omega \to [0,S] \times \Omega, \quad H(s,\omega) = (T(\omega) \wedge s, \omega)$$

is $\mathscr{B}([0,S]) \otimes \mathcal{F}_S$-measurable .

(i)$h_2(s,\omega) = \omega$: This projection map is trivially measurable with respect to $\mathcal{F}_S$.

(ii)$h_1(s,\omega) = T(\omega) \wedge s$:For fixed $s \leq S$, the map $\omega \mapsto T(\omega) \wedge s$ is $\mathcal{F}_s$-measurable (since T is a stopping time). This confirms the process $Z_s = T \wedge s$ is adapted.For fixed ω, the path $s \mapsto T(\omega) \wedge s$ is continuous.A standard result confirms that any adapted process with continuous paths is automatically progressively measurable.Therefore, the map $h_1(s,\omega)$ is $\mathscr{B}([0,S]) \otimes \mathcal{F}_S$-measurable.

Since both components of H are $\mathscr{B}([0,S]) \otimes \mathcal{F}_S$-measurable, H is measurable.

□

Problem 4

Let T be an optional time of $\{\mathcal{F}_t : t \geq 0\}$ on $(\Omega, \mathcal{F}, \mathbb{P})$.

(a) Prove that $\mathcal{F}_{T+}$ is a σ-algebra and $\mathcal{F}_{T+} = \{A \in \mathcal{F} : A \cap \{T < t\} \in \mathcal{F}_t, \forall t \geq 0\}$.

(b) Prove that if T is a stopping time, then $\mathcal{F}_T \subseteq \mathcal{F}_{T+}$.

Proof. (a)Let $\mathcal{G} = \{A \in \mathcal{F} : A \cap \{T < t\} \in \mathcal{F}_t, \forall t \geq 0\}$. Let the standard definition of $\mathcal{F}_{T+}$ be $\mathcal{F}_{T+} = \{A \in \mathcal{F} : A \cap \{T \leq t\} \in \mathcal{F}_{t+}, \forall t \geq 0\}$, where $\mathcal{F}_{t+} = \bigcap_{s>t} \mathcal{F}_s$. The proof consists of two parts: (1) showing $\mathcal{G}$ is a σ-algebra, and (2) showing $\mathcal{G} = \mathcal{F}_{T+}$. For $\mathcal{G}$ is a σ-algebra:

- $\Omega \in \mathcal{G}$: For any $t \geq 0$, $\Omega \cap \{T < t\} = \{T < t\}$. Since T is an optional time, $\{T < t\} \in \mathcal{F}_t$.
- Closure under complement: Let $A \in \mathcal{G}$. Then $A^c \in \mathcal{F}$ and for any $t \geq 0$, $A^c \cap \{T < t\} = \{T < t\} \setminus (A \cap \{T < t\})$. Since $\{T < t\} \in \mathcal{F}_t$ and $A \cap \{T < t\} \in \mathcal{F}_t$, their difference is in $\mathcal{F}_t$. So $A^c \in \mathcal{G}$.
- Closure under countable union: Let $\{A_n\} \subset \mathcal{G}$. Let $A = \bigcup A_n$. Then $A \in \mathcal{F}$. For any $t \geq 0$, $A \cap \{T < t\} = \bigcup (A_n \cap \{T < t\})$. Since each $A_n \cap \{T < t\} \in \mathcal{F}_t$, their countable union is in $\mathcal{F}_t$. So $A \in \mathcal{G}$.

Thus, $\mathcal{G}$ is a σ-algebra.

- $\mathcal{F}_{T+} \subseteq \mathcal{G}$: Consider

$$A \cap \{T < t\} = \bigcup_{n=1}^{\infty} (A \cap \{T \leq t - 1/n\}).$$

- $\mathcal{G} \subseteq \mathcal{F}_{T+}$: Consider

$$A \cap \{T \leq t\} = \bigcap_{n=1}^{\infty} (A \cap \{T \leq t + 1/n\}).$$

Both inclusions hold, therefore $\mathcal{G} = \mathcal{F}_{T+}$.

(b) We just need to verify for any $t \geq 0$,if $A \in \mathcal{F}_T$, the set $A \cap \{T < t\}$ is in $\mathcal{F}_t$.

$$A \cap \{T < t\} = A \cap \left(\bigcup_{n=1}^{\infty} \{T \leq t - 1/n\} \right) = \bigcup_{n=1}^{\infty} (A \cap \{T \leq t - 1/n\}) \in \mathcal{F}_t$$

□

Problem 5

Let $\{X_t : t \geq 0\}$ be a right-continuous, nonnegative supermartingale relative to $\{\mathcal{F}_t : t \geq 0\}$. Prove that $X_\infty = \lim_{t\to\infty} X_t$ exists almost surely and $\{X_t : 0 \leq t \leq \infty\}$ is a supermartingale relative to $\{\mathcal{F}_t : 0 \leq t \leq \infty\}$.

Proof. (1) Take $Y_t = -X_t$ and use submartingale convergence theorem(check $L^1 - bounded$), we can get $Y_\infty = \lim_{t\to\infty} Y_t$ which implies $X_\infty = \lim_{t\to\infty} X_t$.
(2) Define $\mathcal{F}_\infty = \sigma(\cup_{t\geq 0}\mathcal{F}_t)$. We must show $\{X_t : 0 \leq t \leq \infty\}$ is a supermartingale w.r.t. $\{\mathcal{F}_t : 0 \leq t \leq \infty\}$.

(a) Since $X_t \geq 0$ and $X_t \to X_\infty$ a.s., we use Fatou's Lemma:

$$\mathbb{E}[X_\infty] = \mathbb{E}[\liminf_{t\to\infty} X_t] \leq \liminf_{t\to\infty} \mathbb{E}[X_t]$$

Since $\mathbb{E}[X_t]$ is non-increasing in t and non-negative, the limit exists and $\liminf \mathbb{E}[X_t] \leq \mathbb{E}[X_0] < \infty$. Thus, $X_s \in L^1, 0 \leq s \leq \infty$.

(b) X_∞ is the limit of $\mathcal{F}_t$-measurable (and thus $\mathcal{F}_\infty$-measurable) functions, $X_\infty \in \mathcal{F}_\infty$ because σ-algebras are closed under limits.

(c) We only need to check the case $\mathbb{E}[X_\infty|\mathcal{F}_s] \leq X_s$ for $s < \infty$. For any $t > s$, we know $\mathbb{E}[X_t|\mathcal{F}_s] \leq X_s$, which implies $\int_A X_t d\mathbb{P} \leq \int_A X_s d\mathbb{P}$.

Now, apply Fatou's Lemma on the set A:

$$\int_A X_\infty d\mathbb{P} = \int_A \liminf_{t\to\infty} X_t d\mathbb{P} \leq \liminf_{t\to\infty} \int_A X_t d\mathbb{P}$$

Hence

$$\int_A X_\infty d\mathbb{P} \leq \liminf_{t\to\infty} \int_A X_t d\mathbb{P} \leq \int_A X_s d\mathbb{P}.$$

This holds for all $A \in \mathcal{F}_s$, so $\mathbb{E}[X_\infty|\mathcal{F}_s] \leq X_s$.

□

Problem 6

Let $\{X_t : t \geq 0\}$ be a right-continuous, nonnegative submartingale relative to $\{\mathcal{F}_t : t \geq 0\}$.

1. The family $\{X_t : t \geq 0\}$ of r.v.s are uniformly integrable.
2. X_t converges in L^1 as $t \to \infty$.
3. X_t converges almost surely and the limit X_∞ is integrable. Moreover, $\{X_t : 0 \leq t \leq \infty\}$ is a submartingale relative to $\{\mathcal{F}_t : 0 \leq t \leq \infty\}$.

Proof. To ensure the completeness of the proof details, we recommend readers refer to

- Theorem 3.21 in Jean-François Le Gall- Brownian Motion, Martingales, and Stochastic Calculus
- Theorem 9.4.5 in Kai Lai Chung's A Course in Probability Theory
- Theorem 5.5.3 Durrett Probability Theory and Examples(Fourth Edition)

□

Problem 7

Let $\{X_t : t \geq 0\}$ be a right-continuous supermartingale relative to $\{\mathcal{F}_t : t \geq 0\}$ and $S \leq T$ are stopping times of $\{\mathcal{F}_t : t \geq 0\}$. Prove that

1. $\{X_{T\wedge t} : t \geq 0\}$ is supermartingale of $\{\mathcal{F}_t : t \geq 0\}$;
2. $\mathbb{E}[X_{T\wedge t}|\mathcal{F}_S] \leq X_{S\wedge t}$.

Proof. (1)

- By problem 3(b) ,the process $\{X_{T\wedge t} : t \geq 0\}$ is measurable relative to $\{\mathcal{F}_t : t \geq 0\}$.
- Set $Y_t = X_{T\wedge t}$,then $\mathbb{E}(|Y_t|) = \mathbb{E}(Y_t) + 2\mathbb{E}(Y_t^-)$. We can check X_t is supermartingale and $X_t^- = \max\{0, -X_t\}$ is submartingale. By $0 \leq T_t \wedge t \leq t$, we can use OST to get $\mathbb{E}(X_t) \leq \mathbb{E}(X_{T\wedge t}) = \mathbb{E}(Y_t) \leq \mathbb{E}(X_0)$ and $0 \leq \mathbb{E}(X_{T\wedge t}^-) = \mathbb{E}(Y_t^-) \leq \mathbb{E}(X_t^-)$. Since $X_t \in L^1$, we can conclude

$$\mathbb{E}(|X_{T\wedge t}|) = \mathbb{E}(|Y_t|) = \mathbb{E}(Y_t) + 2\mathbb{E}(Y_t^-) < \infty.$$

- By the definition , we must show that for any $0 \leq s < t$, $\mathbb{E}[X_{T\wedge t}|\mathcal{F}_s] \leq X_{T\wedge s}$ a.s.

  On the set $\{T \leq s\}$,the stopping time T occurs at or before time s. Since $s < t$, we have: $T(\omega) \wedge s = T(\omega)$ and $T(\omega) \wedge t = T(\omega)$. Thus, $X_{T\wedge t} - X_{T\wedge s} = X_T - X_T = 0$. The inequality $\mathbb{E}[X_{T\wedge t}|\mathcal{F}_s] \leq X_{T\wedge s}$ holds trivially .

  On the set $\{T > s\}$, we have $T(\omega) \wedge s = s$. Our goal is to prove that on the set $\{T > s\}$, the inequality $\mathbb{E}[X_{T\wedge t}|\mathcal{F}_s] \leq X_s$ holds.

  Define a new process $Z_u = X_{s+u}$ and a new filtration $\mathcal{G}_u = \mathcal{F}_{s+u}$ for $u \geq 0$. Since $\{X_t\}$ is a supermartingale, $\{Z_u\}$ is a supermartingale relative to $\{\mathcal{G}_u\}$. Define a fixed time $v = t - s > 0$ and

a $\mathcal{G}_u$-stopping time $\tau = (T-s)^+$. The stopping time $\tau \wedge v$ is bounded. By the OST applied to Z_u:

$$E[Z_{\tau\wedge v}|\mathcal{G}_0] \leq Z_0$$

Substituting back, $Z_0 = X_s$ and $\mathcal{G}_0 = \mathcal{F}_s$. The inequality becomes:

$$E[Z_{\tau\wedge v}|\mathcal{F}_s] \leq X_s$$

We analyze the variable $Z_{\tau\wedge v} = X_{s+(\tau\wedge v)}$ by cases:

On $\{T \leq s\}$: $\tau = (T-s)^+ = 0$, so $s + (\tau \wedge v) = s$. Thus, $Z_{\tau\wedge v} = X_s$. On $\{T > s\}$: $\tau = T - s$, so $s + (\tau \wedge v) = s + ((T-s) \wedge (t-s)) = \min(T,t) = T \wedge t$. Thus, $Z_{\tau\wedge v} = X_{T\wedge t}$.

Since $Z_{\tau\wedge v} = X_s \mathbf{1}_{\{T\leq s\}} + X_{T\wedge t}\mathbf{1}_{\{T>s\}}$, substitute this into the OST inequality:

$$E[X_s\mathbf{1}_{\{T\leq s\}} + X_{T\wedge t}\mathbf{1}_{\{T>s\}}|\mathcal{F}_s] \leq X_s$$

By linearity and since $X_s\mathbf{1}_{\{T\leq s\}}$ is $\mathcal{F}_s$-measurable

$$X_s\mathbf{1}_{\{T\leq s\}} + E[X_{T\wedge t}\mathbf{1}_{\{T>s\}}|\mathcal{F}_s] \leq X_s$$

Rewrite X_s on the RHS as $X_s(\mathbf{1}_{\{T\leq s\}} + \mathbf{1}_{\{T>s\}})$ and cancel the $X_s\mathbf{1}_{\{T\leq s\}}$ term from both sides:

$$E[X_{T\wedge t}\mathbf{1}_{\{T>s\}}|\mathcal{F}_s] \leq X_s\mathbf{1}_{\{T>s\}}$$

Rearranging the terms gives the desired result:

$$E[(X_{T\wedge t} - X_s)\mathbf{1}_{\{T>s\}}|\mathcal{F}_s] \leq 0$$

(2). Let $t \geq 0$ be fixed. By OST, we have $\mathbb{E}[X_{T\wedge t} \mid \mathcal{F}_{S\wedge t}] \leq X_{S\wedge t}$.

Let $A \in \mathcal{F}_S$ be an arbitrary event. Define $S_A = S$ on A, ∞ on A^c. $T_A = T$ on A, ∞ on A^c. Since $S \leq T$, we have $S_A \leq T_A$. Now apply OST again, we have $S_A \wedge t \leq T_A \wedge t$ and $\mathbb{E}[X_{T_A\wedge t}] \leq \mathbb{E}[X_{S_A\wedge t}]$.

Because

$$\mathbb{E}[X_{S_A\wedge t}] = \int_\Omega X_{S_A\wedge t}dP = \int_A X_{S\wedge t}dP + \int_{A^c} X_t dP$$
$$\mathbb{E}[X_{T_A\wedge t}] = \int_\Omega X_{T_A\wedge t}dP = \int_A X_{T\wedge t}dP + \int_{A^c} X_t dP,$$

the inequality $\mathbb{E}[X_{T_A\wedge t}] \leq \mathbb{E}[X_{S_A\wedge t}]$ becomes

$$\int_A X_{T\wedge t}dP + \int_{A^c} X_t dP \leq \int_A X_{S\wedge t}dP + \int_{A^c} X_t dP$$

Hence $\int_A X_{T\wedge t}dP \leq \int_A X_{S\wedge t}dP$.

This can be written as $\mathbb{E}[\mathbb{I}_A X_{T\wedge t}] \leq \mathbb{E}[\mathbb{I}_A X_{S\wedge t}]$. Since this holds for any arbitrary event $A \in \mathcal{F}_S$, by the definition of conditional expectation,

$$\mathbb{E}[X_{T\wedge t} \mid \mathcal{F}_S] \leq X_{S\wedge t}.$$

□

Problem 8

Let $\{X_t : t \geq 0\}$ be a right-continuous process such that $X_t \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ for $\forall t \geq 0$. Prove that X is a submartingale relative to $\{\mathcal{F}_t : t \geq 0\}$ if and only if $\mathbb{E}[X_T] \geq \mathbb{E}[X_S]$ for all bounded stopping times $S \leq T$ of $\{\mathcal{F}_t : t \geq 0\}$.

Proof.

- Since stopping time S and T are bounded and X is a $\mathcal{F}_t$-submartingale , we can use OST and get $\mathbb{E}(X_T|\mathcal{F}_S) \geq X_S$.Take expection on both sides , then $\mathbb{E}[X_T] \geq \mathbb{E}[X_S]$.
- Since $X_t \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ and X is adapted, we just need to verify $\forall A \in \mathcal{B}(\mathbb{R}^d), \forall t \geq s \geq 0$,

$$\mathbb{E}(X_t 1_A) \geq \mathbb{E}(X_s 1_A).$$

Take $S = s$ and $T(w) = t$ when $w \in A; T(w) = s$ when $w \in A^c$.Then

$$\mathbb{E}[X_T] = \mathbb{E}[X_T 1_A] + \mathbb{E}[X_T 1_{A^c}] = \mathbb{E}[X_t 1_A] + \mathbb{E}[X_s 1_{A^c}]$$
$$\mathbb{E}[X_S] = \mathbb{E}[X_S 1_A] + \mathbb{E}[X_S 1_{A^c}] = \mathbb{E}[X_s 1_A] + \mathbb{E}[X_s 1_{A^c}]$$

Thanks to $\mathbb{E}[X_T] \geq \mathbb{E}[X_S]$, we can get $\mathbb{E}(X_t 1_A) \geq \mathbb{E}(X_s 1_A)$ which implies $\mathbb{E}(X_t|\mathcal{F}_s) \geq X_s$.

□

Problem 9

Let X be a normal Markov process with transition density functions $\{\rho_{s,t} : 0 \leq s \leq t\}$. Prove that

$$\rho_{r,t}(x, z) = \int_{\mathbb{R}^d} \rho_{r,s}(x, y)\rho_{s,t}(y, z)dy$$

for any $0 \leq r \leq s \leq t$ and $x, z \in \mathbb{R}^d$.

Proof. For any $A \in \mathcal{B}(\mathbb{R}^d)$,$\forall 0 \leq r \leq s \leq t$,$\forall x \in \mathbb{R}^d$, we have

$$\begin{aligned}
\int_{\mathbb{R}^d} 1_A(z)\rho_{r,t}(x, z)dz &= T_{r,t}(1_A)(x) = T_{r,s}T_{s,t}(1_A)(x) \\
&= T_{r,s}[T_{s,t}(1_A)(x)] \\
&= T_{r,s}\left(\int_{\mathbb{R}^d} 1_A(z)\rho_{s,t}(x, z)dz\right) := T_{r,s}(g)(x) \\
&= \int_{\mathbb{R}^d} g(y)\rho_{r,s}(x, y)dy \\
&= \int_{\mathbb{R}^d}\int_{\mathbb{R}^d} 1_A(z)\rho_{s,t}(y, z)\rho_{r,s}(x, y)dydz \quad \text{(by Fubini's Theorem)} \\
&= \int_{\mathbb{R}^d} 1_A(z)\left(\int_{\mathbb{R}^d} \rho_{r,s}(x, y)\rho_{s,t}(y, z)dy\right)dz.
\end{aligned}$$

Since A is arbitrary, we get $\rho_{r,t}(x, z) = \int_{\mathbb{R}^d} \rho_{r,s}(x, y)\rho_{s,t}(y, z)dy$. □

2 Brownian motion

Problem 1

Let $X = \{X_t, t \geq 0\}$ be a stochastic process with independent increments. Prove that, for all $0 \leq s < t$, the increment $X_t - X_s$ is independent of $\mathcal{F}_s^X = \sigma(\{X_u : 0 \leq u \leq s\})$.

Proof. Let $Y = X_t - X_s$. We aim to prove that $\sigma(Y)$ is independent of $\mathcal{F}_s^X$.

We define a π-system $\mathcal{P}$ which generates $\mathcal{F}_s^X$:

$$\mathcal{P} = \{A : A = \{(X_{u_1}, X_{u_2}, \ldots, X_{u_n}) \in D\}, \text{ where } 0 \leq u_1 < \cdots < u_n \leq s, D \in \mathcal{B}(\mathbb{R}^n), n \geq 1\}.$$

Let $A \in \mathcal{P}$ and $B = \{Y \in E\} \in \sigma(Y)$, where $E \in \mathcal{B}(\mathbb{R}^n)$. We have

$$A \in \sigma(\{X_0, X_{u_1} - X_0, X_{u_2} - X_{u_1}, \ldots, X_{u_n} - X_{u_{n-1}}\})$$

where $0 \leq u_1 < \cdots < u_n \leq s$.

Since Y is independent of all variables in $\{X_0, X_{u_1} - X_0, X_{u_2} - X_{u_1}, \ldots, X_{u_n} - X_{u_{n-1}}\}$, it follows that A and B are independent events, i.e. $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. This shows that $\sigma(Y)$ is independent of the π-system $\mathcal{P}$.

We define $\mathcal{L} = \{A \subset \Omega : A \text{ is independent of } Y\}$. We can check $\mathcal{L}$ is a λ-system . It satisfies:

1. $\Omega \in \mathcal{L}$.
2. If $A \in \mathcal{L}$, then $A^c \in \mathcal{L}$.
3. If $\{A_i\}_{i=1}^{\infty} \subset \mathcal{L}$ are disjoint, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{L}$.

Proof of 3. Let $\{A_i\}_{i=1}^{\infty} \subset \mathcal{L}$ be a sequence of pairwise disjoint sets. Let $A = \bigcup_{i=1}^{\infty} A_i$.For any $B \in \sigma(Y)$, we have:

$$\mathbb{P}(A \cap B) = \mathbb{P}\left(\left(\bigcup_{i=1}^{\infty} A_i\right) \cap B\right) = \mathbb{P}\left(\bigcup_{i=1}^{\infty} (A_i \cap B)\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)\mathbb{P}(B) = \mathbb{P}(B) \sum_{i=1}^{\infty} \mathbb{P}(A_i) = \mathbb{P}(B)\mathbb{P}(A)$$

where we use the sequence of sets $\{A_i \cap B\}$ is also disjoint,and $A_i \in \mathcal{L}$. Therefore, A is independent of Y, and $A \in \mathcal{L}$.

From before, we have shown that $\mathcal{P} \subset \mathcal{L}$. The π-λ Theorem states: If $\mathcal{P}$ is a π-system and $\mathcal{L}$ is a λ-system such that $\mathcal{P} \subset \mathcal{L}$, then the smallest σ-algebra containing $\mathcal{P}$ must also be contained in $\mathcal{L}$. Hence

$$\sigma(\mathcal{P}) \subset \mathcal{L} \to \mathcal{F}_s^X \subset \mathcal{L}$$

This means every set $A \in \mathcal{F}_s^X$ is also in $\mathcal{L}$, and thus, A is independent of $Y = X_t - X_s$.Therefore, $\mathcal{F}_s^X$ is independent of $\sigma(X_t - X_s)$. $\square$

Problem 2 1. Prove that a standard Brownian motion is a time-homogeneous Markov process.

2. Determine the family of Markov evolution operators.

3. Determine whether a standard Brownian motion is a (strong) Feller process or not.

4. Determine the infinitesimal generator of standard Brownian motion.

Proof. (1) Let $B = \{B_t : t \geq 0\}$ be a standard Brownian motion adapted to a filtration $\{\mathcal{F}_t\}_{t\geq 0}$. Fix $s, t \geq 0$ and let $f : \mathbb{R} \to \mathbb{R}$ be a bounded, Borel-measurable function.

Using the property of conditional expectation (specifically, if X is $\mathcal{F}$-measurable and Y is independent of $\mathcal{F}$, then $\mathbb{E}[g(X, Y) \mid \mathcal{F}] = h(X)$ where $h(x) = \mathbb{E}[g(x, Y)]$), we obtain:

$$\mathbb{E}[f(B_{t+s}) \mid \mathcal{F}_s] = \mathbb{E}[f(B_s + (B_{t+s} - B_s)) \mid \mathcal{F}_s] = g(B_s),$$

where the function $g(x) = \mathbb{E}[f(x + (B_{t+s} - B_s))] = \int_{-\infty}^{\infty} f(x + y) \frac{1}{\sqrt{2\pi t}} e^{-\frac{y^2}{2t}}\, dy$.

Since $\mathbb{E}[f(B_{t+s}) \mid \mathcal{F}_s]$ is a function of B_s alone, the Markov property holds:

$$\mathbb{E}[f(B_{t+s}) \mid \mathcal{F}_s] = \mathbb{E}[f(B_{t+s}) \mid B_s].$$

Furthermore, the transition probability function $P_{s,s+t}(x, A) = P(B_{t+s} \in A \mid B_s = x)$ is determined by the density:

$$p(t, x, z) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{(z-x)^2}{2t}}.$$

This transition density depends only on the time difference $(s + t) - s = t$. Therefore, the process is time-homogeneous.

(2) For any $f \in \mathcal{B}_b(\mathbb{R})$ and $x \in \mathbb{R}$, the operator $T_{s,t}$ is defined by the conditional expectation:

$$T_{s,t} f(x) = \mathbb{E}[f(B_t) \mid B_s = x] = \int_{-\infty}^{\infty} f(y) \frac{1}{\sqrt{2\pi(t-s)}} e^{-\frac{(y-x)^2}{2(t-s)}}\, dy.$$

(3) Standard Brownian motion is a Feller process, and furthermore, it is a Strong Feller process.

Part 1: Proof that Brownian Motion is a Feller Process.

Let $C_0(\mathbb{R})$ be the space of continuous functions vanishing at infinity, equipped with the supremum norm $\|f\|_\infty = \sup_{x\in\mathbb{R}} |f(x)|$. The transition semigroup of standard Brownian motion is given by:

$$P_t f(x) = \mathbb{E}[f(x + B_t)] = \int_{\mathbb{R}} f(y) p(t, x, y)\, dy = \int_{\mathbb{R}} f(x + z) g_t(z)\, dz,$$

where $g_t(z) = \frac{1}{\sqrt{2\pi t}} e^{-z^2/2t}$ is the heat kernel.

(a) **P_t maps $C_0(\mathbb{R})$ into $C_0(\mathbb{R})$:** Let $f \in C_0(\mathbb{R})$.

- Continuity: Since f is continuous and bounded, and g_t is integrable, by the Dominated Convergence Theorem, $x \mapsto P_t f(x)$ is continuous.

- Vanishing at infinity: Since $f \in C_0(\mathbb{R})$, for any $\epsilon > 0$, there exists K such that $|f(y)| < \epsilon$ for $|y| > K$.

$$|P_t f(x)| \leq \int_{\mathbb{R}} |f(x+z)| g_t(z) dz.$$

As $|x| \to \infty$, the mass of the integral is concentrated in regions where f is small. Thus, $\lim_{|x|\to\infty} P_t f(x) = 0$.

(b) **Continuity at $t = 0$:** We need to show $\lim_{t\to 0^+} \|P_t f - f\|_\infty = 0$ for any $f \in C_0(\mathbb{R})$. Since f is uniformly continuous on $\mathbb{R}$ (because $f \in C_0$), for any $\epsilon > 0$, there exists $\delta > 0$ such that $|y| < \delta \implies |f(x+y) - f(x)| < \epsilon$ for all x.

$$|P_t f(x) - f(x)| = \left| \int_{\mathbb{R}} (f(x+z) - f(x)) g_t(z) dz \right| \leq \int_{\mathbb{R}} |f(x+z) - f(x)| g_t(z) dz.$$

Split the integral into $|z| < \delta$ and $|z| \geq \delta$. As $t \to 0$, the mass of the Gaussian kernel concentrates on $|z| < \delta$, making the integral arbitrarily small uniformly in x. Thus, Brownian motion is a **Feller process**.

Part 2: Proof that Brownian Motion is a Strong Feller Process.

A Markov process is *Strong Feller* if P_t maps bounded measurable functions $B_b(\mathbb{R})$ into bounded continuous functions $C_b(\mathbb{R})$ for any $t > 0$.

Let $f \in B_b(\mathbb{R})$ (measurable and bounded, i.e., $|f(x)| \leq M$). Consider the function $P_t f(x)$:

$$P_t f(x) = \int_{\mathbb{R}} f(y) \frac{1}{\sqrt{2\pi t}} e^{-\frac{(y-x)^2}{2t}} dy.$$

Since the Gaussian density $p(t, x, y)$ is infinitely differentiable with respect to x for any $t > 0$, and f is bounded, we can apply the Dominated Convergence Theorem to show continuity. Specifically, for any $x_n \to x$:

$$\lim_{n\to\infty} \int_{\mathbb{R}} f(y) p(t, x_n, y) dy = \int_{\mathbb{R}} f(y) p(t, x, y) dy.$$

The limit can be taken inside the integral because $|f(y)p(t, x_n, y)|$ is dominated by an integrable function (since $p(t, \cdot, \cdot)$ is bounded and decays rapidly). Thus, $x \mapsto P_t f(x)$ is continuous.

(4) Let $B_t = (B_t^1, \ldots, B_t^d)$ be a standard Brownian motion in $\mathbb{R}^d$ starting at x. For $f \in C_b^2(\mathbb{R}^d)$, the infinitesimal generator L is defined as:

$$Lf(x) = \lim_{t\to 0^+} \frac{\mathbb{E}^x[f(B_t)] - f(x)}{t}.$$

Using the multivariate Taylor expansion of $f(B_t)$ around x up to the second order:

$$f(B_t) - f(x) = \sum_{i=1}^{d} \frac{\partial f}{\partial x_i}(x)(B_t^i - x_i) + \frac{1}{2} \sum_{i,j=1}^{d} \frac{\partial^2 f}{\partial x_i \partial x_j}(x)(B_t^i - x_i)(B_t^j - x_j) + o(\|B_t - x\|^2).$$

Taking the expectation $\mathbb{E}^x[\cdot]$, and applying the properties of standard Brownian motion:

$$\mathbb{E}^x[B_t^i - x_i] = 0,$$

$$\mathbb{E}^x[(B_t^i - x_i)(B_t^j - x_j)] = t\delta_{ij} = \begin{cases} t & i = j \\ 0 & i \neq j \end{cases}.$$

The expectation of the expansion becomes:

$$\mathbb{E}^x[f(B_t)] - f(x) = 0 + \frac{1}{2}\sum_{i=1}^{d} \frac{\partial^2 f}{\partial x_i^2}(x) \cdot t + o(t).$$

Substituting this back into the limit:

$$Lf(x) = \lim_{t \to 0^+} \frac{\frac{t}{2}\sum_{i=1}^{d} \frac{\partial^2 f}{\partial x_i^2}(x) + o(t)}{t} = \frac{1}{2}\sum_{i=1}^{d} \frac{\partial^2 f}{\partial x_i^2}(x).$$

Thus, the generator is:

$$L = \frac{1}{2}\Delta,$$

where Δ is the Laplace operator.

□

Problem 3

Let B be a Brownian motion on $(\Omega, \mathcal{F}, \mathbb{P})$ relative to some filtration $\{\mathcal{F}_t : t \geq 0\}$. Prove that there exists $F \in \mathcal{F}$ with $\mathbb{P}(F) = 1$, such that

$$F \subset \{\omega \in \Omega : \forall t \geq 0, \text{ either } D^- B_t(\omega) = +\infty \text{ or } D_- B_t(\omega) = -\infty\}$$

Proof. It's suffices to show with probability one, Brownian paths are not Lipschitz continuous.See **Durrett 4th,Theorem 8.1.6.**

Fix a constant $C < \infty$ and let $A_n = \{\omega : \text{there is an } s \in [0,1] \text{ so that } |B_t - B_s| \leq C|t-s| \text{ when } |t-s| \leq 3/n\}$. For $1 \leq k \leq n-2$, let

$$Y_{k,n} = \max\left\{\left|B\left(\frac{k+j}{n}\right) - B\left(\frac{k+j-1}{n}\right)\right| : j = 0,1,2\right\}$$

$$B_n = \{\text{at least one } Y_{k,n} \leq 5C/n\}$$

The triangle inequality implies $A_n \subset B_n$. The worst case is $s = 1$. We pick $k = n-2$ and observe

$$\begin{aligned}\left|B\left(\frac{n-3}{n}\right) - B\left(\frac{n-2}{n}\right)\right| &\leq \left|B\left(\frac{n-3}{n}\right) - B(1)\right| + \left|B(1) - B\left(\frac{n-2}{n}\right)\right| \\ &\leq C(3/n + 2/n)\end{aligned}$$

Using $A_n \subset B_n$ and the scaling relation (8.1.1) now gives

$$\begin{aligned} P(A_n) \le P(B_n) \le nP(|B(1/n)| \le 5C/n)^3 &= nP(|B(1)| \le 5C/n^{1/2})^3 \\ &\le n\{(10C/n^{1/2}) \cdot (2\pi)^{-1/2}\}^3 \end{aligned}$$

since $\exp(-x^2/2) \le 1$. Letting $n \to \infty$ shows $P(A_n) \to 0$. Noticing $n \to A_n$ is increasing shows $P(A_n) = 0$ for all n and completes the proof.

□

Problem 4

Prove that the sample paths of a standard 1-dimensional Brownian motion are nowhere Hölder-continuous with Hölder order $\gamma > 1/2$ a.s.

Proof. From **Brownian Motion,Peter Mörters and Yuval Peres.**
It suffices to show that, for fixed $\varepsilon > 0$ and $c > 0$, almost surely, for all $t \ge 0$, there exists $0 < h < \varepsilon$ with $|B(t+h) - B(t)| > ch^\alpha$. By Brownian scaling we may further assume $\varepsilon = 1$. Note that, after this simplification, the complementary event means that there is a $t_0 \ge 0$ such that

$$\sup_{h\in(0,1)} \frac{B(t_0+h) - B(t_0)}{h^\alpha} \le c \quad \text{or} \quad \inf_{h\in(0,1)} \frac{B(t_0+h) - B(t_0)}{h^\alpha} \ge -c.$$

We may assume that $t_0 \in [0,1)$. Fix $l \ge 1/(\alpha - 1/2)$. Then $t_0 \in [\frac{k-1}{2^n}, \frac{k}{2^n})$ for any large n and some $0 \le k < 2^n - l$. Then, by the triangle inequality, for all $j \in \{1, \dots, 2^n - k\}$,

$$\left| B\left(\frac{k+j}{2^n}\right) - B\left(\frac{k+j-1}{2^n}\right) \right| \le 2c \left(\frac{j+1}{2^n}\right)^\alpha.$$

Now, for any $0 \le k < 2^n - l$, let $\Omega_{n,k}$ be the event

$$\left\{ \left| B\left(\frac{k+j}{2^n}\right) - B\left(\frac{k+j-1}{2^n}\right) \right| \le 2c \left(\frac{l+1}{2^n}\right)^\alpha \quad \text{for } j = 1, 2, \dots, l \right\}.$$

It suffices to show that, almost surely for all sufficiently large n and all $k \in \{0, \dots, 2^n - l\}$ the event $\Omega_{n,k}$ does not occur. Observe that

$$\mathbb{P}(\Omega_{n,k}) \le \left[\mathbb{P}\left(|B(1)| \le 2^{n/2} 2c \left(\frac{l+1}{2^n}\right)^\alpha \right) \right]^l \le \left[2^{n/2} 2c \left(\frac{l+1}{2^n}\right)^\alpha \right]^l,$$

since the normal density is bounded by $1/2$. Hence, for a suitable constant C,

$$\mathbb{P}\left(\bigcup_{k=0}^{2^n - l} \Omega_{n,k} \right) \le 2^n \left[2^{n/2} 2c \left(\frac{l+1}{2^n}\right)^\alpha \right]^l = C \left[2^{1-l(\alpha-1/2)} \right]^n,$$

which is summable. Thus

$$\mathbb{P}\left(\limsup_{n\to\infty} \bigcup_{k=0}^{2^n - l} \Omega_{n,k} \right) = 0.$$

This is the required statement and hence the proof is complete. □